The Recursive Observability Filter: A Dynamical-Systems Framework for Civilizational Detectability and the Fermi Paradox

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July 30, 2026

Abstract

The Fermi paradox is usually posed as a tension between the abundance of cosmic opportunity and the absence of confirmed evidence for extraterrestrial technological civilizations. Most quantitative treatments vary *how many* civilizations arise while holding detectability fixed. We argue that detectability itself should be a dynamical variable, and develop the *Recursive Observability Filter*: a compact three-variable dynamical system coupling an effective capability I, a regulatory capacity R, and an external observability O, embedded in an explicit decomposition $N_{\text{obs}} = N_{\text{true}}\, P_{\text{surv}}\, h(O)\, P_{\text{search}}$ that separates existence, signal production and search coverage. We prove two structural properties of the formulation: observability is bounded below by a thermodynamic floor, and regulatory capacity is confined to the unit interval, both by construction rather than by clamping. A consequence is that peak-then-decline observability — the signature usually *assumed* in low-observability accounts of the Great Silence — instead *emerges* whenever signal suppression scales faster in capability than signal production. We treat the Lambert-W threshold that motivated this work as an explicitly auxiliary diagnostic, and show that the threshold governing the integrated system is logarithmic rather than Lambert-type. The model is implemented twice, independently, in Python and in JavaScript; the two agree to within solver tolerance across twenty configurations, and the formulation is checked against closed-form solutions, an independent high-order integrator, and property-based testing over 300 randomised parameter sets. We claim no solution to the Fermi paradox: the contribution is a transparent, falsifiable modelling environment in which the consequences of assumptions about recursive intelligence, regulation and observability can be examined and, where they fail, seen to fail.

Keywords: Fermi paradox; Drake equation; technosignatures; detectability; recursive intelligence; nonlinear dynamics; Lambert W; astrobiology; model verification

1 Introduction

1.1 The paradox is an inference problem

The universe is old, planets are common, and many planetary systems predate the Solar System by billions of years (Lineweaver, 2001; Borucki, 2016). Yet no extraterrestrial signal, probe, artefact or unambiguous technosignature has been confirmed. The tension between these statements is conventionally called the Fermi paradox, and is usually compressed into the question "where is everybody?" (Hart, 1975; Brin, 1983; Webb, 2015).

Stated that way the question is misleading, because it treats non-observation as evidence about existence. It is not. Non-observation constrains the *product* of several quantities: how many civilizations arise, how many survive to a technologically active phase, how strong a signature they produce, and whether our searches have covered the region of parameter space in which that signature would appear (Gray, 2015; Tarter, 2001; Sheikh, 2020). A more careful formulation asks which assumptions would make extraterrestrial technological civilizations visible *to us, now*.

This distinction is not merely rhetorical. It determines what a model of the paradox should vary. Most quantitative treatments — from the Drake equation onward — vary the number of civilizations while holding detectability fixed, typically inside a single lifetime parameter L (Prantzos, 2013, 2020). The alternative pursued here is to give detectability its own dynamics.

1.2 Existence, detectability, observation

We distinguish three quantities that are frequently conflated:

$$N_{\text{true}} \;\geq\; N_{\text{detectable}} \;\geq\; N_{\text{obs}}, \tag{1}$$

the number of civilizations that exist, the number producing signatures detectable in principle, and the number actually detected by our searches. A civilization may exist without being detectable; it may be detectable in principle without ever being observed, if our search has not intersected its signature. [ESTABLISHED]

The framework developed here modifies the middle term. It asks how a civilization's external observability evolves as its capability, regulatory capacity, energy use, expansion behaviour and information efficiency change — and in particular what happens across a transition to recursive, self-improving technological capability.

1.3 Why recursive capability is the interesting transition

A civilization becomes qualitatively different when its tools improve the process of making better tools. Automated discovery, engineering and design close a loop between capability and the rate of capability growth (Good, 1965; Bostrom, 2014; Russell, 2019). Quantum computation enters this picture in exactly this way: not as a separate phenomenon, but as one possible contributor to the rate at which a civilization can recursively improve its own technology, by accelerating the simulation, search and optimisation problems that gate technological progress (Shor, 1994; Nielsen and Chuang, 2010; Preskill, 2018). [SPECULATIVE]

Such a loop plausibly changes more than growth rate. It may change information efficiency, and hence signal leakage; it may change whether expansion is attractive; it may shorten or lengthen the interval over which a civilization is loud. None of this is asserted here as fact. The model asks what *follows* if such feedback occurs, and makes the dependence on that assumption explicit.

1.4 What breaks if observability is modelled carelessly

Giving observability its own differential equation is easy to do badly, and the failure mode is instructive enough that we report it. If the terms representing compression, efficiency and deliberate quietness are written as an absolute subtraction — a flux removed from O at a rate that does not depend on O — then nothing prevents observability from passing through zero and running to large negative values. A trajectory can then be classified as "quiet" on the strength of a physically meaningless number.

That is not a coding slip; it is a modelling error with a clear physical diagnosis. Compression and stealth reduce the signal a civilization *emits*. They are fractional reductions, and one cannot suppress emissions that are not being made. Writing them as fractional rates acting on the emitted signal, together with an explicit floor representing irreducible waste heat, makes observability bounded below by construction. We prove this as Proposition 2 and give the analogous statement for regulatory capacity as Proposition 1.

The correction has a substantive consequence, which is the main modelling result of this paper. Once suppression is fractional, the quasi-steady observability is a *ratio* of production to suppression. Peak-then-decline behaviour then follows automatically whenever suppression grows faster in capability than production does — it no longer has to be imposed by selecting a peaked functional form. A low-observability account of the Great Silence that assumes a peaked $O(I)$ assumes its conclusion; one that derives the peak from the ratio of two independently motivated mechanisms does not.

1.5 Contributions

1. An observability-aware decomposition $N_{\text{obs}} = N_{\text{true}}\, P_{\text{surv}}\, h(O)\, P_{\text{search}}$ that separates survival, signal production and search coverage, and locates precisely where a dynamical detectability model acts on the Drake equation (§3).
2. A three-variable dynamical model of capability, regulation and observability derived from mechanism balances, with two structural guarantees — $O \geq O_{\text{floor}}$ and $R \in [0, 1]$ — that hold by construction rather than by clamping (§4).
3. Peak-then-decline observability as a *derived* consequence of suppression outgrowing production, rather than an assumed functional form (§7).
4. A registry architecture in which every functional form is interchangeable, making the question "do the conclusions survive changing the functions?" a menu selection rather than a rewrite (§4).
5. A verification protocol unusual in this literature: closed-form solutions, an independent high-order integrator, property-based testing over randomised parameter sets, and two independently written implementations agreeing to solver tolerance (§6).
6. An explicitly negative result about the Lambert-W threshold that motivated this work: the balance that justifies W is not the balance governing the integrated system (§5).

1.6 What this paper does not claim

It does not claim that extraterrestrial civilizations exist, that advanced civilizations become invisible, that recursive artificial intelligence causes collapse or transcendence, or that the Fermi paradox is solved. Every substantive statement carries a strength label — [ESTABLISHED], [PLAUSIBLE], [HYPOTHETICAL] or [SPECULATIVE] — and these are collected in Table 6.

2 Background

2.1 The Drake equation and what exoplanet science settled

The Drake equation organises the expected number of communicative civilizations as a product,

$$N = R_* f_p n_e f_l f_i f_c L, \tag{2}$$

and is best read as a map of uncertainty rather than a calculator (Drake and Sobel, 1992). Exoplanet surveys have substantially constrained its leading astrophysical factors: planets are common and potentially habitable planets are no longer speculative (Borucki, 2016; Batalha, 2014; Dressing and Charbonneau, 2015; Kopparapu et al., 2013; Seager, 2013). [ESTABLISHED]

The consequence is that uncertainty has migrated to the right of Eq. (2): to the emergence of life and intelligence, and above all to L, the interval over which a civilization is detectable.

Prantzos (2020) shows explicitly that the L factor dominates the inference. Our framework acts on precisely this term, by refusing to treat it as a scalar.

2.2 Great Filter and existential risk

The Great Filter frames the problem as a bottleneck: some step between non-living matter and a long-lived expansive civilization must be extremely improbable (Hanson, 1998). Whether that step is behind or ahead of us carries obvious stakes, and the literature on existential risk explores the latter possibility in detail (Bostrom, 2002, 2013; Ord, 2020; Häggström, 2016). Ćirković (2018) surveys the philosophical terrain, and Verendel and Häggström (2017) give a Bayesian treatment.

The Great Filter is a statement about probability mass, not about dynamics. It does not, by itself, say how a civilization moves through the bottleneck, nor what it looks like from outside while doing so. That is the gap a dynamical model can address.

2.3 Technosignatures and the search

What could actually be detected has been studied since Cocconi and Morrison (1959) and Dyson (1960). Energy-use signatures scale with the Kardashev classification (Kardashev, 1964); infrared searches for large-scale energy capture have produced null results at survey scale (Wright et al., 2014; Griffith et al., 2015; Garrett, 2015); and the field has increasingly formalised how to reason about coverage of the search space (Tarter, 2001; Wright, 2018; Sheikh, 2020; Wright et al., 2022).

Two points matter here. First, different signature classes correspond to different civilizational behaviours — waste heat, narrowband beacons, propulsion, megastructures, atmospheric markers — so "detectability" is not one quantity. Second, the searches conducted so far cover a small fraction of the parameter space, which is why P_{search} must appear explicitly in any honest inference (Wright et al., 2022). [ESTABLISHED]

2.4 Expansion, percolation and timing

If interstellar settlement is feasible, the galaxy should be crossed quickly on astronomical timescales (Hart, 1975; Tipler, 1980; Armstrong and Sandberg, 2013), which sharpens the paradox considerably. Percolation and agent-based treatments soften it by allowing settlement to stall (Landis, 1998; Forgan, 2009). Anthropic and hard-steps arguments address why we find ourselves early (Carter, 1983; Watson, 2008; Kipping, 2020), and the grabby-aliens analysis of Hanson et al. (2021) makes the sharp observation that if loud civilizations explain human earliness, then quiet ones must also be rare.

That last result is the most direct challenge to any low-observability account of the Great Silence, including ours, and we return to it in §8.

2.5 Recursive intelligence

The possibility that capability improves the machinery of capability has been discussed since Good (1965), and elaborated with attention to control and alignment by Bostrom (2014), Russell (2019), Omohundro (2008) and Yudkowsky (2008). Quantum computation is relevant to this framework only through this channel — as one possible accelerant of the recursive improvement loop (Shor, 1994; Preskill, 2018; Nielsen and Chuang, 2010) — and we are explicit that its magnitude and even its sign of effect are unconstrained. [SPECULATIVE]

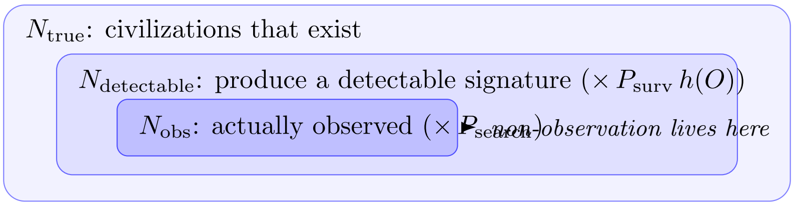

Figure 1: Existence, detectability and observation are distinct. Non-observation constrains the innermost set, and therefore only the *product* of the factors in Eq. (3); it does not by itself constrain N_{true}.

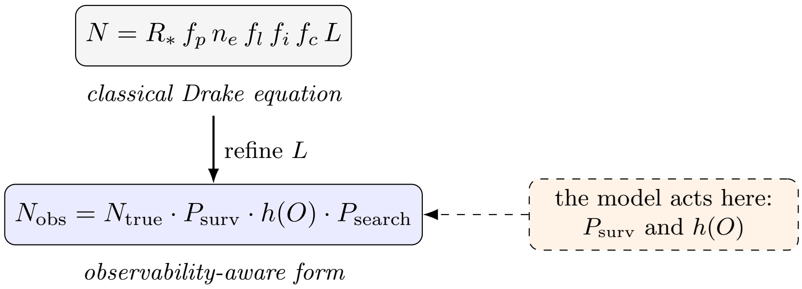

Figure 2: The framework is a refinement of the Drake equation's lifetime and detectability content, not a replacement for it. It leaves the astrophysical factors untouched and acts on survival and signal production.

2.6 The gap

Across these literatures, detectability is treated as a static property: a coefficient, a lifetime, or a survey limit. Where it is allowed to change, the change is typically imposed as a chosen profile. What is missing is a formulation in which observability has its own equation of motion, is coupled to the same processes that drive capability and regulation, and is constrained to remain physical. Supplying that, and testing it seriously, is the purpose of this paper.

3 Separating existence, detectability and observation

3.1 The decomposition

Let N_{true} be the number of technological civilizations that exist in the relevant volume and epoch. Write P_{surv} for the probability that such a civilization persists into the phase we could in principle detect, $h(O)$ for the probability that it produces a signature strong enough to register given its external observability O, and P_{search} for the probability that our searches intersect that signature. Then

$$\boxed{N_{\text{obs}} \;=\; N_{\text{true}} \cdot P_{\text{surv}} \cdot h(O) \cdot P_{\text{search}}} \tag{3}$$

with h a saturating map from signature strength to detection probability. We take

$$h(O) \;=\; 1 - e^{-\kappa O}, \tag{4}$$

where κ sets instrument sensitivity, and define the detection probability for a single civilization as $P_{\text{det}} = h(O)\, P_{\text{search}}$.

Equation (3) is a bookkeeping identity, not a theory; its value is that it forces the four mechanisms apart. [ESTABLISHED] Figure 1 shows the resulting nesting and Figure 2 shows where the framework acts relative to Eq. (2).

3.2 Why this matters for inference

Suppose a survey reports no detections. Under Eq. (3) the likelihood of that outcome depends on P_{search} and on the distribution of O across the population, not on N_{true} alone. Two very different worlds — one in which civilizations are rare, and one in which they are common but mostly quiet during the epoch we can sample — produce the same null result. Distinguishing them requires a model of how O is distributed and how it evolves, which is what §4 supplies. [ESTABLISHED]

3.3 What observability is, and is not

We use O for *external observability*: the strength of signatures leaving the system, aggregating waste heat, leakage and intentional emission, expansion footprint, and engineered structures. It is not energy use, and it is not capability. A civilization may be extremely capable and comparatively quiet; it may also be modest and loud.

Crucially, O can be small but never zero. Any energy use produces thermodynamic consequences, and a claim of true invisibility would require physics we have no warrant to assume (Dyson, 1960; Kardashev, 1964). We encode this directly: O is bounded below by a strictly positive floor O_{floor} (Proposition 2). Low observability in this framework always means "hard to detect", never "absent". [ESTABLISHED]

4 The dynamical model

4.1 Variables

The model evolves three coupled quantities in a dimensionless time τ.

$I(\tau)$ — **effective capability.** A compressed systems variable aggregating scientific capability, automation, computational capacity, coordination and technological leverage. It is emphatically *not* a measure of intelligence in the psychometric sense, nor of moral worth, nor of consciousness. A civilization may grow in I without becoming wiser, safer or louder.

$R(\tau)$ — **regulatory capacity.** Alignment, governance, safety culture, institutional learning, epistemic reliability and error correction: the capacity to prevent runaway or destructive instability. We treat it as a *quality index* normalised to $[0, 1]$, which is what makes it commensurable with the growth rates it opposes (§4.3).

$O(\tau)$ — **external observability.** Signature strength as defined in §3, bounded below by $O_{\text{floor}} > 0$.

Table 1 collects these with their drivers. Figure 3 maps the conventional civilizational stages onto them and Figure 4 shows the coupling structure.

4.2 Capability

Capability grows by ordinary accumulation, aAI, and by recursive amplification, $bA_{\text{rec}}F(I)$, where F encodes how aggressive the recursion is. It is damped by regulatory friction, cRI, and saturates through physical, energetic and coordination limits, $s_I I^2$:

$$\frac{\mathrm{d}I}{\mathrm{d}\tau} = aAI + bA_{\text{rec}}F(I) - cRI - s_I I^2. \tag{5}$$

Each term carries a mechanism, which is the point: the equation is assembled from interpretable balances rather than written down for convenience.

Table 1: Model variables and drivers. Drivers are exogenous inputs; the three state variables are integrated.

Symbol	Role	Interpretation
I	state	effective capability: science, automation, computation, coordination
R	state	regulatory quality index in $[0, 1]$: alignment, governance, error correction
O	state	external observability: leakage, waste heat, expansion footprint
A	driver	ordinary amplification pressure (baseline technological leverage)
A_{rec}	driver	recursive amplification: self-improvement, automated design, compute scaling
A_{ref}	driver	reflective amplification: capability directed at safety and foresight
Q	driver	institutional quality

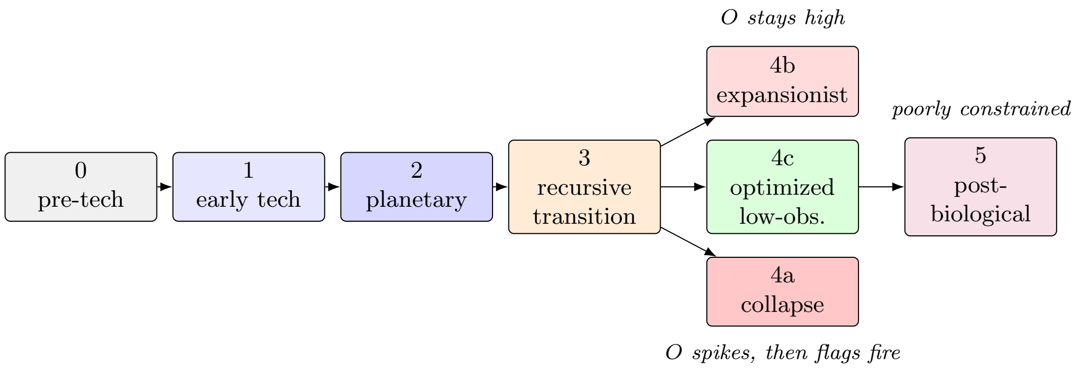

Figure 3: Civilizational stages and the branch structure at the recursive transition. Stages 0–2 are pre-recursive; the branch at stage 3 into expansionist, optimized-quiet and collapse outcomes is where the framework does its work.

4.3 Regulation

Regulatory capacity is built by reflective capability and institutional quality, and eroded by the pace of recursive amplification. The essential modelling choice is that erosion must be proportional to the regulation that exists — one cannot erode capacity that is absent — and that building acts on the remaining headroom:

$$\frac{\mathrm{d}R}{\mathrm{d}\tau} = \underbrace{(uA_{\mathrm{ref}} + vQ)}_{B}(1 - R) - \underbrace{(wA_{\mathrm{rec}} + s_R)}_{W} R. \tag{6}$$

Proposition 1 (Regulation is confined to the unit interval)**.** *If $B \geq 0$ and $W \geq 0$ and $R(0) \in [0, 1]$, then $R(\tau) \in [0, 1]$ for all $\tau > 0$, and*

$$R_\infty = \frac{B}{B + W} = \frac{uA_{\mathrm{ref}} + vQ}{uA_{\mathrm{ref}} + vQ + wA_{\mathrm{rec}} + s_R}. \tag{7}$$

Proof. At $R = 0$, Eq. (6) gives $\mathrm{d}R/\mathrm{d}\tau = B \geq 0$, so the lower boundary cannot be crossed downwards. At $R = 1$ it gives $\mathrm{d}R/\mathrm{d}\tau = -W \leq 0$, so the upper boundary cannot be crossed upwards. Equation (6) is linear in R with constant coefficients when the drivers are constant, $\mathrm{d}R/\mathrm{d}\tau = B - (B + W)R$, whence the stated fixed point and $R(\tau) = R_\infty + (R_0 - R_\infty)e^{-(B+W)\tau}$. $\square$

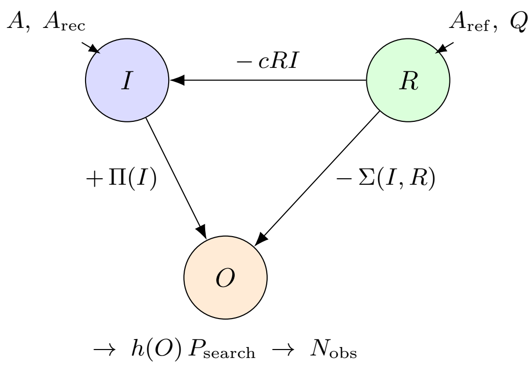

Figure 4: Coupling structure. Regulation damps capability through $-cRI$; observability is produced by capability, $\Pi(I)$, and suppressed fractionally by capability *and* regulation together, $\Sigma(I, R)$. Note the deliberate absence of an arrow from I to R: in Eq. (6) regulatory capacity responds only to the exogenous drivers A_{ref} and Q and to erosion by A_{rec}, not to capability itself. Closing that loop is a natural extension (§9).

Two consequences deserve emphasis. First, R_∞ is a *share* — the fraction of total pressure that is constructive — which makes it directly interpretable. Second, because $R \geq 0$, the damping term $-cRI$ in Eq. (5) can never change sign. A formulation permitting $R < 0$ would make failing governance *accelerate* capability, which is not a mechanism anyone intends to model.

4.4 Observability

Observability is produced by energy use, broadcast leakage and expansion footprint, and suppressed by compression and stealth. Suppression is fractional: it acts on the signal being emitted, above an irreducible floor.

$$\frac{\mathrm{d}O}{\mathrm{d}\tau} = \underbrace{pE(I) + qB_{\mathrm{c}}(I) + rX(I)}_{\Pi(I)} - (O - O_{\mathrm{floor}})\underbrace{\big(mC(I,R) + nS(I,R)\big)}_{\Sigma(I,R)}. \tag{8}$$

Proposition 2 (Observability is bounded below by the floor)**.** *If* $\Pi \geq 0$ *and* $\Sigma \geq 0$ *and* $O(0) \geq O_{\mathrm{floor}}$*, then* $O(\tau) \geq O_{\mathrm{floor}}$ *for all* $\tau > 0$*. Where* $\Sigma > 0$*, the quasi-steady observability is*

$$O^* = O_{\mathrm{floor}} + \frac{\Pi(I)}{\Sigma(I,R)}. \tag{9}$$

Proof. At $O = O_{\mathrm{floor}}$ the suppression term vanishes identically and $\mathrm{d}O/\mathrm{d}\tau = \Pi \geq 0$, so the floor cannot be crossed downwards. Setting $\mathrm{d}O/\mathrm{d}\tau = 0$ in Eq. (8) and solving for O gives Eq. (9). □

Equation (9) is the analytical heart of the paper. Because O^* is a *ratio*, its behaviour in I is governed by the relative scaling of production and suppression, not by either alone:

$$\Pi \sim I, \quad \Sigma \sim IR \implies O^* \sim \frac{1}{R} \qquad \text{(observability independent of capability)}, \tag{10}$$

$$\Pi \sim I, \quad \Sigma \sim I^2R \implies O^* \sim \frac{1}{IR} \qquad \text{(observability declines with capability)}. \tag{11}$$

The second case produces peak-then-decline without any peaked functional form being selected. That is the sense in which the signature is derived rather than assumed, and we exhibit it numerically in §7. [PLAUSIBLE]

Table 2: Interchangeable functional forms. Any combination is admissible; the solver never references a specific choice.

Component	Variants	Form
$F(I)$ recursion	linear, superlinear, Lambert, exponential, saturating	I; I^2; Ie^I; e^I; $I/(1+I/K)$
$O(I)$ static	A increasing, B decreasing, C peaked, D threshold	$1-e^{-\lambda I}$; $e^{-\lambda I}$; $Ie^{-\lambda I}$; $\sigma\big(k(I-I_c)\big)$
E energy use	linear, Kardashev	I; I^2
B_c broadcast	linear, decaying	I; $Ie^{-\lambda I}$
X expansion	linear, inward	I; 0
C, S suppression	linear, superlinear	IR; I^2R

Table 3: Regime criteria, in evaluation order. Thresholds are model parameters, not empirical quantities.

#	Regime	Criterion	Strength
1	collapse proxy	runaway *and* ($R < R_{\min}$ or O peaked then fell below $O_{\text{detectable}}$)	[HYPOTHETICAL]
2	runaway	$I > I_{\text{runaway}}$ or $A_{\text{rec}}/(R+\epsilon) > \Theta$	[SPECULATIVE]
3	pre-detectable	$\max O < O_{\text{detectable}}$ and $\max I < I_{\text{advanced}}$	[PLAUSIBLE]
4	optimized low-observable	$I \geq I_{\text{advanced}}$, $R \geq R_{\min}$, $O < O_{\text{detectable}}$	[HYPOTHETICAL]
5	expansionist visible	$I \geq I_{\text{advanced}}$ and $O \geq O_{\text{detectable}}$	[HYPOTHETICAL]
6	visible technological	$P_{\text{det}} > 0.01$	[PLAUSIBLE]
7	uncertain	none of the above	[PLAUSIBLE]

4.5 Interchangeable functional forms

No functional form above is privileged. Each is drawn from a registry (Table 2), so that testing whether a conclusion survives a change of functions is a menu selection rather than a rewrite. This makes the robustness requirement operational rather than aspirational.

4.6 Regime classification

Trajectories are labelled by the criteria in Table 3, evaluated in a fixed priority order. The ordering is not commutative: a trajectory meeting two criteria receives the earlier label, and we state this explicitly because it affects interpretation of the phase portrait in §7.

Regime 1 is named a *proxy* deliberately. The model contains no mortality term, so it registers the conditions under which collapse is expected without integrating a collapse trajectory; we return to this in §9.

5 The Lambert threshold, and why it is auxiliary

This section reports a negative result. The Lambert W function motivated the original formulation of this work, and it is mathematically correct where we use it — but the balance that justifies W is *not* the balance governing the system we integrate. We set this out in full rather than presenting the boundary as though it were a property of the dynamics, because the distinction is exactly the kind of thing that separates a usable model from decorative mathematics.

5.1 When W is actually required

The Lambert W function is defined implicitly by $W(y)e^{W(y)} = y$, and is required precisely when an unknown appears both inside and outside an exponential (Corless et al., 1996). A relation of the form

$$xe^x = y \quad \Longrightarrow \quad x = W(y) \tag{12}$$

genuinely needs it; a relation of the form $e^x = y$ does not, being solved by a logarithm. Any use of W that does not rest on the former structure is ornamental.

5.2 The natural balance does not need W

Take the Lambert recursion kernel $F(I) = Ie^I$ from Table 2. Equation (5) becomes

$$\frac{\mathrm{d}I}{\mathrm{d}\tau} = aAI + bA_{\mathrm{rec}}Ie^I - cRI - s_I I^2. \tag{13}$$

Setting recursive amplification against regulatory damping, and temporarily neglecting ordinary growth and saturation, gives

$$bA_{\mathrm{rec}}Ie^I \;=\; cRI. \tag{14}$$

For $I > 0$ the factor I *cancels on both sides*, leaving

$$bA_{\mathrm{rec}}e^I = cR \qquad \Longrightarrow \qquad I_{\mathrm{thr}} = \ln\!\left(\frac{cR}{bA_{\mathrm{rec}}}\right), \tag{15}$$

a *logarithmic* threshold. The Lambert structure is destroyed by the very cancellation that the Ie^I kernel appears to guarantee. This is the central observation of this section, and it is easy to miss.

5.3 Recovering Lambert structure requires a different balance

Lambert structure survives only if Ie^I is compared against a regulatory capacity that is *not* itself proportional to I. Introducing such a capacity,

$$C_R \;=\; c\,R^\eta\,\ln(1 + A_{\mathrm{ref}})\,\mathcal{E}, \tag{16}$$

with η a regulation exponent and $\mathcal{E}$ an alignment factor, and imposing $bA_{\mathrm{rec}}Ie^I = C_R$, yields

$$Ie^I = \frac{C_R}{bA_{\mathrm{rec}}} \qquad \Longrightarrow \qquad \boxed{I_{\mathrm{crit}} \;=\; W\!\left(\frac{c\,R^\eta\,\ln(1 + A_{\mathrm{ref}})\,\mathcal{E}}{bA_{\mathrm{rec}}}\right).} \tag{17}$$

Equation (17) is a clean Lambert boundary, and it is what Figure 5 displays.

5.4 The honest reading

C_R appears nowhere in Eq. (5). The system we actually integrate contains the damping term $-cRI$, whose threshold is Eq. (15) — logarithmic. Equation (17) therefore describes a balance we have *posited alongside* the dynamics, not one the dynamics contain.

We therefore state plainly:

> The Lambert boundary $I_{\mathrm{crit}} = W(C_R/bA_{\mathrm{rec}})$ is an *auxiliary stability diagnostic* evaluated under a stated auxiliary balance. It is not a bifurcation of the integrated system, and no result in §7 depends on it. [HYPOTHETICAL]

The implementation enforces this distinction rather than relying on the text: the Lambert branch is guarded so that W is evaluated only when the Lambert kernel is actually selected, and every other kernel falls back to the logarithmic threshold of Eq. (15). Verification of the identity $I_{\mathrm{crit}}e^{I_{\mathrm{crit}}} = C_R/(bA_{\mathrm{rec}})$ to machine precision is reported in §6.

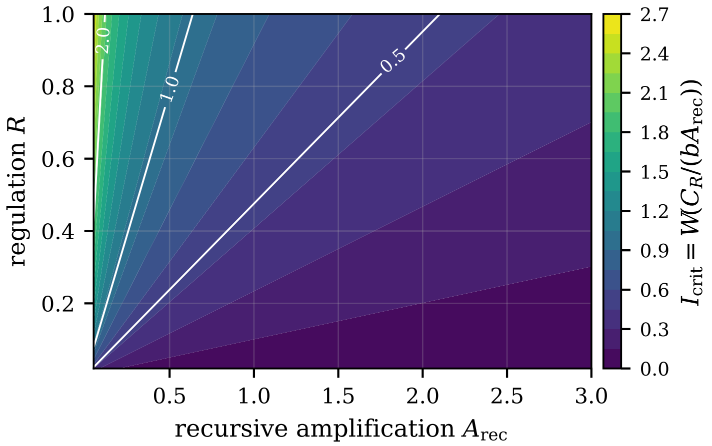

Figure 5: The auxiliary Lambert boundary I_{crit} over the (A_{rec}, R) plane, from Eq. (17). Higher regulatory capacity raises the critical capability that can be absorbed; stronger recursive amplification lowers it. This surface is a diagnostic under the auxiliary balance of Eq. (16), not a bifurcation set of the integrated system.

5.5 Why keep it at all

Three reasons. It provides an analytic handle on the intuition that regulation can absorb only so much recursive amplification; it makes the comparison between regulatory capacity and amplification pressure explicit and plottable; and, reported honestly, it is a worked example of the discipline the framework demands elsewhere — a candidate piece of mathematics tested for whether it is required, and found to be required only under an assumption we then have to declare.

6 Numerical methods and verification

Models of this kind are rarely verified beyond "the code runs and the plots look plausible". Because the conclusions here depend on the qualitative shape of computed trajectories, we treat verification as a first-class part of the contribution and report it in full.

6.1 Integration

The system (5)–(8) is integrated with an explicit adaptive Dormand–Prince Runge–Kutta method of order 5(4) with dense output (Dormand and Prince, 1980), as provided by SciPy (Virtanen et al., 2020; Harris et al., 2020), at relative tolerance 10^{-6} and absolute tolerance 10^{-9}. All exponentials are argument-clipped to $[-50, 50]$ to prevent overflow.

Two of the recursion kernels in Table 2 — superlinear and exponential — produce genuine finite-time blow-up, where I diverges at a finite τ. An adaptive stepper approaching such a singularity takes ever-smaller steps and does not terminate. Integration therefore stops on a terminal event when I exceeds a bound I_{abort} set two orders of magnitude above the runaway threshold, and the remaining output samples hold the terminal state. We note that padding instead with the last *output* sample is misleading: the singularity generally falls between samples, so the recorded maximum can understate a divergence by orders of magnitude.

6.2 Six independent oracles

Each check below tests the implementation against something other than itself. Results are summarised in Table 4.

Table 4: Verification results. Each oracle tests the implementation against something other than itself.

	Oracle	Scope	Result
A	closed-form $R(\tau)$	19 configurations	max error 2.3×10^{-6}
B	independent 8th-order integrator	20 configurations	20/20 agree
C	quadrature of $\mathrm{d}O/\mathrm{d}\tau$	12 non-singular configurations	$\leq 1.5 \times 10^{-4}$
D	Lambert identity	2304 evaluations	4.4×10^{-16}
E	structural invariants	300 random configurations	**0 violations**
F	Python vs. JavaScript	20 configurations	$\lvert\Delta I\rvert = 5.2 \times 10^{-5}$; labels identical

A. Closed-form regulation. Equation (6) is linear in R and — importantly — independent of I and O. It therefore integrates exactly, giving $R(\tau) = R_\infty + (R_0 - R_\infty)e^{-(B+W)\tau}$ with R_∞ from Eq. (7). Every configuration reproduces this to $\leq 2.3 \times 10^{-6}$, consistent with the requested tolerance.

B. Independent high-order integrator. Every configuration was re-integrated with an eighth-order Dormand–Prince method at tolerances $10^{-12}/10^{-14}$ and compared against the production solver at its default settings. All twenty agree. For the two blow-up kernels the comparison is restricted to the interval before the abort event, since the reference has no such guard.

C. Independent quadrature of the observability equation. $\mathrm{d}O/\mathrm{d}\tau$ was reconstructed from the computed I, R and integrated by quadrature, independently of the ODE solver's treatment of O. Agreement is $\leq 1.5 \times 10^{-4}$ on all non-singular configurations.

D. The Lambert identity. I_{crit} must satisfy its own definition, $I_{\text{crit}}e^{I_{\text{crit}}} = C_R/(bA_{\text{rec}})$. Across 576 parameter combinations and four values of R — 2304 evaluations — the maximum relative residual is 4.4×10^{-16}, i.e. machine precision. The validity guard also behaves as specified, returning no boundary for any non-Lambert kernel.

E. Property-based invariant testing. Propositions 1 and 2 assert structural bounds, so they should hold for *arbitrary* admissible parameters rather than for tuned presets. Three hundred randomised parameter sets were drawn across the full declared ranges, with randomised recursion kernels and suppression forms. There were **no violations**: the global minimum observability was 2.16×10^{-3}, above the floor, and regulation remained within $[4.0 \times 10^{-3},\ 0.999]$. This is the strongest single piece of evidence that the bounds are structural and not artefacts of parameter choice.

F. Two independent implementations. The model is implemented twice: once in Python against SciPy, and once in JavaScript with a separately written adaptive Dormand–Prince integrator, its own dense-output interpolation, its own initial-step heuristic, and a hand-implemented Lambert W_0 via Halley iteration. Across the twenty configurations the worst absolute discrepancies are $\lvert\Delta I\rvert = 5.2 \times 10^{-5}$, $\lvert\Delta R\rvert = 3.1 \times 10^{-7}$ and $\lvert\Delta O\rvert = 2.7 \times 10^{-5}$; I_{crit} agrees to 2.2×10^{-12} relative; and *all twenty regime classifications are identical*. Figure 6 shows the comparison with residuals.

The independent W_0 implementation was itself checked against the defining identity over $x \in [10^{-6}, 10^{6}]$ and across the negative branch $x \in [-1/e, 0)$, with maximum relative residual 1.0×10^{-15}.

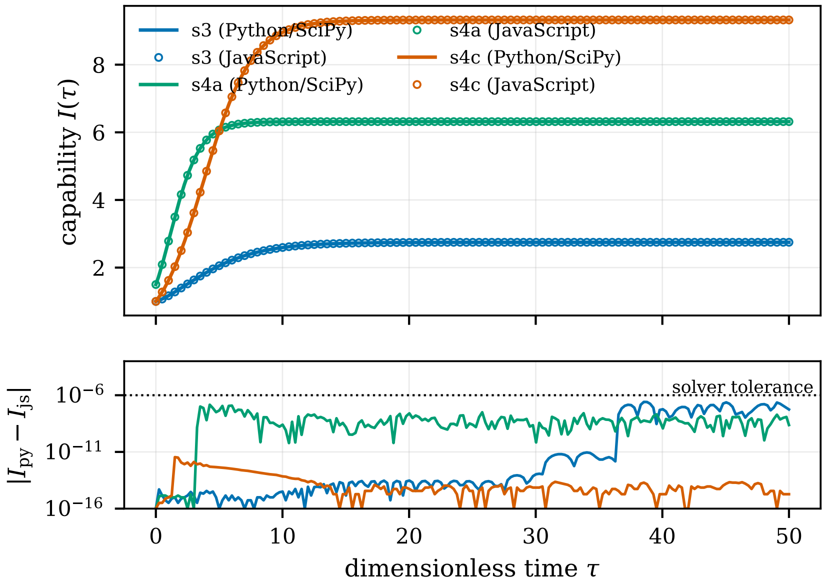

Figure 6: Cross-validation of two independent implementations. Upper: capability trajectories for three stages, Python (lines) and JavaScript (markers). Lower: absolute difference, which stays at or below the requested solver tolerance across the integration. The two codes share no numerical machinery.

6.3 What verification does and does not establish

These checks establish that the equations we describe are the equations being solved, and that they are solved accurately. They say nothing about whether the equations describe a real civilization: no term here has been calibrated against data, because no relevant data exist. The distinction between a verified model and a validated one is worth keeping sharp, and we make no claim to the latter. [ESTABLISHED]

7 Results

All results below use the parameter set in Appendix C and are reproducible from the accompanying code.

7.1 Stage trajectories

Figure 7 shows capability, regulation and observability for the eight canonical stage configurations. Table 5 compares the outcome of each against the behaviour that stage is *defined* to exhibit — a consistency check we regard as a minimum standard, and one that a model can easily fail without anyone noticing.

Two configurations are worth reading closely. Stage 1, the early technological phase, reduces to a logistic capability equation with no regulatory damping, and reaches its analytic limits exactly: $I \to aA/s_I = 3.0$ and $O \to 1 - e^{-\lambda I} = 0.5934$ (Appendix B). Stage 4a drives regulation to $R_\infty = 0.0438$, below the collapse threshold $R_{\min} = 0.05$, *without R becoming negative* — collapse proceeds by the intended mechanism of regulatory exhaustion rather than by a sign inversion in the damping term.

7.2 Peak-then-decline is derived, not imposed

This is the central result. Figure 8 contrasts two formulations of the observability equation on the same stage-4c configuration.

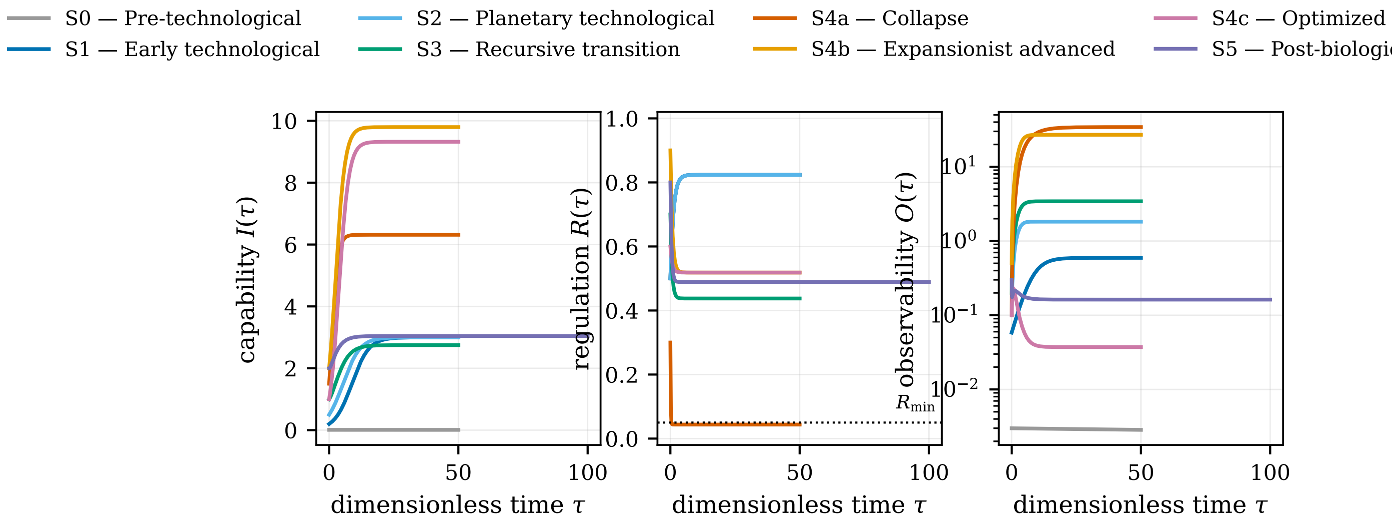

Figure 7: Capability, regulation and observability for the eight stage configurations. Regulation is confined to $[0, 1]$ by Proposition 1; observability is shown logarithmically and remains strictly positive by Proposition 2, spanning almost five orders of magnitude across stages.

Table 5: Specified versus simulated behaviour for each stage configuration.

Stage	Specified behaviour	Simulated	Regime
0	$\mathrm{d}I/\mathrm{d}\tau \approx 0$, no signature	$I = 0.010$, $O = 0.003$	pre-detectable
1	$I \to aA/s_I$, O rising	$I = 3.0000$, $O = 0.5934$	visible technological
2	capability grows, O still rising	$I = 3.000$, $O : 0.10 \to 1.82$	visible technological
3	full model, recursion active	$I = 2.75$, $O = 3.43$	visible technological
4a	regulation collapses	$R \to 0.0438 < R_{\min}$, $R \geq 0$	collapse proxy
4b	high I, high O sustained	$I = 9.80$, $O = 27.0$	expansionist visible
4c	O peaks, then declines	$0.100 \to 0.231 \to 0.037$	optimized low-observable
5	speculative	$I = 3.04$, $O : 0.30 \to 0.16$	visible technological

Panel (a) uses an absolute suppression term, $-mC - nS$, whose rate does not depend on O. Observability passes through zero at $\tau \approx 0.3$ and falls without bound, reaching -1431 by the end of the integration. Under the criteria of Table 3 this trajectory is still labelled "optimized low-observable", because the classifier reads $O < O_{\text{detectable}}$. The label is arithmetically correct and physically vacuous: the civilization is not quiet, it is off the bottom of the model.

Panel (b) uses the fractional form of Eq. (8). Observability rises from 0.100 to a peak of 0.231 at $\tau = 0.65$, then decays to 0.037 — below the detection threshold, above the thermodynamic floor, and strictly positive throughout.

The important point is *why* it declines. Nothing in this configuration selects a peaked functional form; O is integrated, not evaluated. The decline follows from Eq. (11): compression and stealth scale as I^2R while production scales as I, so the quasi-steady observability $O^* \sim 1/(IR)$ falls as capability grows. A civilization becomes harder to see because efficiency outpaces output, which is a mechanism rather than a stipulation. [PLAUSIBLE]

This distinction matters for the standing objection to low-observability accounts of the Great Silence — that they assume what they set out to explain. A model that selects a peaked $O(I)$ does assume it. One that obtains the peak from the ratio of two independently motivated terms does not, and can be falsified by showing that the required scaling separation is implausible.

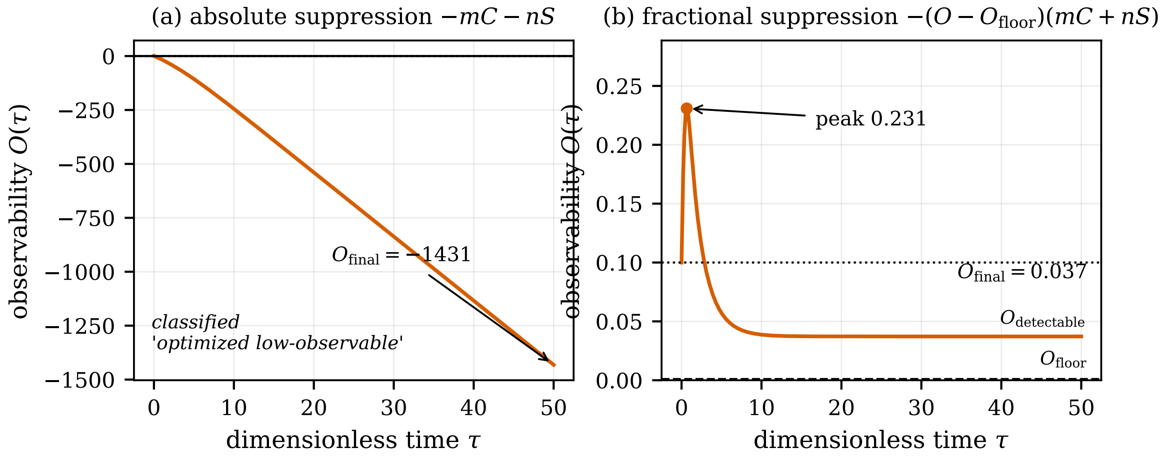

Figure 8: Why the form of the suppression term matters. (a) With absolute suppression, observability passes through zero and diverges negatively, yet is still classified "optimized low-observable". (b) With fractional suppression and a thermodynamic floor, the same configuration produces a genuine peak-then-decline, bounded below — and the decline is derived from suppression scaling faster in capability than production, not imposed by choosing a peaked profile.

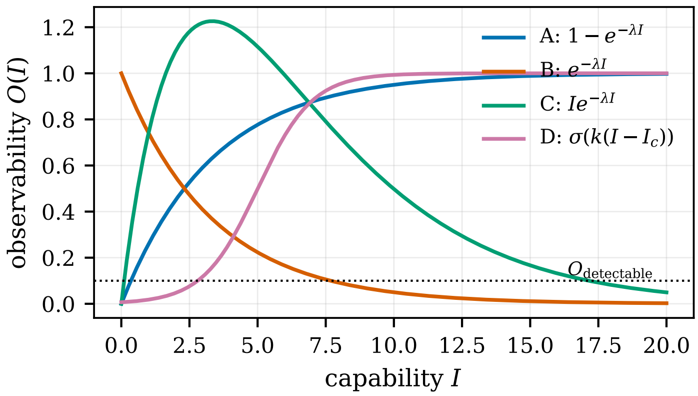

Figure 9: The four interchangeable static observability models. Conclusions about the Great Silence must be reported together with the model that produced them; the low-observability regime does not arise under model A.

7.3 Robustness across observability models

Figure 9 shows the four static observability models of Table 2. These span monotone-increasing, monotone-decreasing, peaked and threshold behaviour, and the framework admits all of them.

The relevant robustness question is not which curve is "right" but whether a conclusion survives changing it. The low-observability regime is reachable under the peaked and decreasing models and under the dynamical form of Eq. (8); it is *not* reachable under the increasing model, which instead produces the expansionist-visible regime seen in stage 4b. We regard this as the correct behaviour: a framework in which every functional choice yielded the same conclusion would not be testing anything. [ESTABLISHED]

7.4 Phase structure

Figure 10 shows the trajectories in the (I, R) plane. Because regulation is confined to $[0, 1]$, the phase space is a bounded strip and the regime boundaries are directly readable: the vertical line at $I_{\rm advanced}$ separates pre-advanced from advanced capability, and the horizontal line at $R_{\rm min}$ marks regulatory exhaustion.

The branch structure of Figure 3 is visible here. Configurations sharing an initial condition

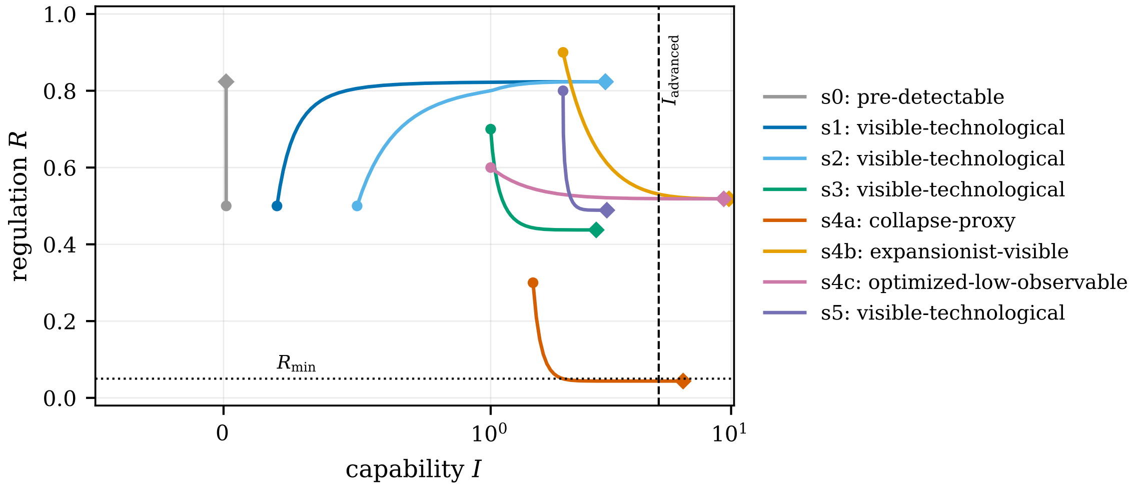

Figure 10: Phase portrait in the capability–regulation plane; circles mark initial conditions and diamonds final states. Regulation is bounded to $[0, 1]$, so the reachable phase space is a strip rather than a half-plane, and the collapse boundary R_{min} is approached but never crossed into negative values.

separate according to whether regulation is sustained: those retaining $R \gtrsim 0.5$ move right along the strip into the advanced regimes, while stage 4a falls to the lower boundary and triggers the collapse proxy. [HYPOTHETICAL]

7.5 Decomposing the silence

Finally, Eq. (3) lets us ask which factor a null result actually constrains. Taking stage 4c as an illustration, with $O = 0.037$, $\kappa = 1$ and $P_{\text{search}} = 0.5$, the single-civilization detection probability is

$$P_{\text{det}} = \left(1 - e^{-\kappa O}\right) P_{\text{search}} = 0.018, \tag{18}$$

against $P_{\text{det}} = 0.50$ for the expansionist stage 4b at $O = 27$. The same underlying population is therefore some 27 times less likely to be detected in the quiet regime than in the loud one, with no change in N_{true} and no extinction event. [HYPOTHETICAL]

We stress what this does *not* show. It does not show that civilizations are quiet, nor that quietness explains the Great Silence. It shows that under this framework a null result is consistent with a wide range of N_{true}, and that attributing non-detection to rarity requires an argument about O and P_{search} that is usually left implicit.

8 Discussion

8.1 Relation to existing solution classes

The framework is not a rival to the standard families of Fermi-paradox explanations but a way of expressing several of them in a common state space.

Great Filter. A filter is a statement about where probability mass is lost. In this framework a filter ahead of us appears as a region of parameter space from which trajectories reach the collapse proxy — and the model makes explicit that a filter acting on *observability* rather than survival is also possible, and would be empirically indistinguishable from a survival filter using non-detection alone (Hanson, 1998; Ćirković, 2018). [HYPOTHETICAL]

Grabby aliens. Hanson et al. (2021) argue that if loud, expanding civilizations explain human earliness, then quiet civilizations must also be rare. This is the sharpest challenge to any low-observability account, ours included, and we do not claim to answer it. What the framework contributes is a place to state the disagreement precisely: the grabby-aliens argument constrains the *joint* distribution of expansion behaviour and timing, which in our terms is a constraint on the population distribution of the expansion term X and the branch probabilities at stage 3 — quantities the present model takes as given rather than derives. Coupling the two is the obvious next step.

Zoo and non-interference hypotheses. These correspond to a stealth term S that is large and deliberate (Ball, 1973; Bracewell, 1960). The framework accommodates them but cannot distinguish deliberate quietness from emergent efficiency, since both enter Σ identically. That degeneracy is worth stating: behavioural and thermodynamic explanations of quietness are not separable by observability alone.

Percolation and expansion limits. Models in which settlement stalls (Landis, 1998; Forgan, 2009; Armstrong and Sandberg, 2013) act on X, the expansion contribution to production, and are compatible with the "inward" variant in Table 2.

8.2 Thermodynamic discipline

The floor O_{floor} is not a numerical convenience. Any civilization using energy radiates, and the possibility of true invisibility would require assumptions we have no basis to make (Dyson, 1960; Kardashev, 1964). Building the floor into the equation rather than into the prose has a practical benefit: it makes the strong claim unavailable. One cannot accidentally conclude from this model that a civilization vanished, only that it became hard to detect at a stated sensitivity κ and coverage P_{search}. [ESTABLISHED]

This is the substantive difference between the corrected and uncorrected formulations of §7.2. The uncorrected version permitted arbitrarily negative observability, and therefore permitted — indeed produced — a conclusion that the framework's own thermodynamic commitments forbid.

8.3 Implications for search strategy

If the low-observability regime is populated at all, it has a testable consequence. In that regime production is dominated by residual energy use rather than by broadcast or expansion, since it is precisely the directed and efficient channels that suppression acts on most strongly. Search programmes sensitive to waste heat and infrared excess would then retain sensitivity where narrowband beacon searches lose it (Dyson, 1960; Wright et al., 2014; Garrett, 2015).

This is a conditional, not a recommendation: it holds if C2 holds, and C2 is labelled [HYPOTHETICAL]. Its value is that it is the kind of statement that could be wrong. Sustained null results from infrared surveys at improving sensitivity would push against it (Griffith et al., 2015; Wright et al., 2022).

8.4 On the role of recursive and quantum capability

The model treats A_{rec} as an exogenous driver and does not explain where recursive capability comes from. Quantum computation enters only as one possible contributor to that driver — a mechanism by which the simulation, search and optimisation problems gating technological progress might be accelerated (Shor, 1994; Preskill, 2018). We attach no magnitude to this and note that its sign of effect on *observability* is not determined: faster capability growth raises production through E and I, but also raises suppression through C and S, and Eq. (9) shows

Table 6: Claim-strength summary. [ESTABLISHED]: supported by observation or standard theory. [PLAUSIBLE]: compatible with known theory. [HYPOTHETICAL]: a model assumption or model output. [SPECULATIVE]: weakly constrained.

Claim	Strength
Existence, detectability and observation are distinct; non-observation constrains their product	[ESTABLISHED]
Planets are common; astrophysical Drake factors are comparatively well constrained	[ESTABLISHED]
Search coverage P_{search} is small and must appear explicitly in any inference	[ESTABLISHED]
Energy use implies a non-zero signature floor; true invisibility is unavailable	[ESTABLISHED]
Regulation confined to $[0, 1]$ and observability bounded below (Props. 1, 2)	[ESTABLISHED]
Peak-then-decline follows when suppression outscales production	[PLAUSIBLE]
Detectability depends on civilizational behaviour and technology, not only on existence	[PLAUSIBLE]
A high-capability, low-observability regime is reachable under plausible parameters	[HYPOTHETICAL]
Quiet regimes favour waste-heat searches over narrowband beacon searches	[HYPOTHETICAL]
The Lambert boundary is a useful auxiliary diagnostic	[HYPOTHETICAL]
Recursive capability transitions change observability trajectories	[HYPOTHETICAL]
Quantum computation materially accelerates recursive self-improvement	[SPECULATIVE]
All stable high-capability civilizations become quiet	[SPECULATIVE]

that only the ratio matters. A civilization that computes more efficiently may be louder or quieter depending on which term scales faster. [SPECULATIVE]

8.5 Claim strengths

Table 6 collects every substantive claim with its label. We regard this as a necessary discipline in a subject where the distance between a model result and a statement about the universe is easily lost.

9 Limitations and failure criteria

9.1 Limitations

No calibration. Not one coefficient in Eqs. (5)–(8) is constrained by data, because no relevant data exist. The model is dimensionless and exploratory. It supports statements of the form "if these mechanisms operate with these relative strengths, then this follows"; it does not support point predictions, and the numerical values in §7 should be read as illustrations of qualitative regimes rather than as estimates.

Collapse is a proxy, not a trajectory. The model contains no mortality term. Stage 4a therefore raises the collapse criteria — regulation falling below $R_{\min}$, amplification exceeding regulation by more than Θ — without capability actually declining. The "transient spike then disappearance" signature often associated with civilizational collapse is consequently *not* reproduced dynamically. Adding a term coupling mortality to regulatory exhaustion is the most obvious extension.

Stiffness. Making suppression fractional (Eq. (8)) has a numerical cost: the relaxation rate Σ scales as I^2R under the superlinear variants, and an explicit method requires step sizes of order $1/\Sigma$. Integration cost therefore grows linearly with the suppression weights, which bounds how far into the strongly-suppressing corner of parameter space the present solver can economically explore. An implicit or automatically stiffness-switching integrator would remove this, at the cost of the exact numerical parity with the browser implementation reported in §6.

The Lambert boundary is auxiliary. As established in §5, C_R does not appear in the integrated system. No result in this paper depends on the Lambert boundary, and it should not be read as a bifurcation of the dynamics.

Degeneracy of quietness. Deliberate stealth and emergent efficiency enter Σ identically and cannot be distinguished by observability alone.

No feedback from capability to regulation. Equation (6) makes regulatory capacity depend on the exogenous drivers and on erosion by A_{rec}, but not on I itself (Figure 4). This is a real simplification: a highly capable civilization would plausibly be better at building governance as well as at eroding it, and the sign of that net effect is one of the more interesting open questions the framework could address. Closing the loop — for example by making the building term depend on $A_{\mathrm{ref}}I$ — is straightforward and would introduce the possibility of genuine bistability between well-regulated and poorly-regulated high-capability states.

A single civilization, not a population. The model integrates one trajectory. Statements about N_{obs} therefore assume a population whose members share parameters, which is plainly false. A population-level treatment would integrate over distributions of the drivers and branch probabilities, and is required before the framework can be compared quantitatively with grabby-aliens style arguments (Hanson et al., 2021; Prantzos, 2020).

9.2 Failure criteria

We state in advance what would count against the framework, since a model of this kind is otherwise difficult to argue with.

1. **Non-robustness.** If the low-observability regime arises only for a narrow band of suppression scalings, or only under one choice from Table 2, the conclusion is an artefact of that choice. §7 reports the dependence explicitly; the regime does not arise under observability model A.
2. **Assumed rather than derived decline.** If peak-then-decline could be obtained only by selecting a peaked $O(I)$, the framework would add nothing to existing low-observability arguments. Eq. (11) and §7.2 are the response; if the required scaling separation (Σ growing faster in I than Π) can be shown to be physically implausible, that response fails.
3. **Collapse and quietness not separated.** If the model could not distinguish a civilization that ended from one that became quiet, it would reproduce the ambiguity it was built to dissolve. The regime criteria separate them, but only as far as the collapse proxy is meaningful — see above.

4. **Thermodynamic violation.** Any result requiring $O \leq 0$ invalidates the physical interpretation. Proposition 2 makes this unreachable by construction, and the property-based tests of §6 check it empirically.

5. **Search coverage ignored.** If P_{search} is dropped, non-detection is misread as evidence about N_{true}. It appears explicitly in Eq. (3).

6. **Empirical pressure.** Sustained null results from infrared and waste-heat surveys at improving sensitivity would argue against the quiet regime being populated, since that is the channel it predicts remains open (Griffith et al., 2015; Garrett, 2015; Wright et al., 2022).

Criterion 6 is the only one that is empirical rather than internal. That asymmetry is itself a limitation of the subject, not of the model, and we prefer to name it than to disguise it.

10 Conclusions

The Fermi paradox is usually modelled by varying how many civilizations arise. We have argued that it should also be modelled by varying how long they remain externally observable, and that observability is best treated as a state variable with its own equation of motion rather than as a coefficient.

Three results follow.

First, once existence, signal production and search coverage are separated in Eq. (3), non-observation constrains their product and not N_{true} alone. This is a logical point, but it is routinely elided, and it determines what a model of the paradox should vary. [ESTABLISHED]

Second, an observability equation must be constructed so that it cannot leave the physical range. Writing suppression as a fractional rate acting on emitted signal, above an explicit thermodynamic floor, achieves this by construction (Proposition 2); the analogous treatment of erosion confines regulatory capacity to the unit interval (Proposition 1). These are not cosmetic. The uncorrected formulation produced, and confidently labelled, a "quiet" civilization whose observability was -1431.

Third — and this is the modelling result we would defend most strongly — peak-then-decline observability need not be assumed. Because the quasi-steady observability is a ratio of production to suppression, the decline emerges whenever suppression scales faster in capability than production does (Eq. (11)). A low-observability account of the Great Silence built this way does not assume its conclusion, and can be attacked on the plausibility of a scaling separation rather than on the choice of a curve. [PLAUSIBLE]

We have also reported a negative result. The Lambert W threshold that named and motivated this work is not a property of the integrated system: the natural balance cancels I and yields a logarithm, and recovering Lambert structure requires an auxiliary balance we must declare rather than derive (§5). We keep the boundary as a diagnostic and depend on it nowhere.

Finally, the framework is verified rather than merely implemented. Closed-form solutions, an independent high-order integrator, property-based testing over 300 randomised parameter sets, and two independently written implementations agreeing to solver tolerance together establish that the equations described here are the equations solved. They establish nothing about whether those equations describe a real civilization, and we have been careful to keep verification and validation apart.

The framework does not solve the Fermi paradox and is not intended to. Its purpose is to make a class of arguments about cosmic silence precise enough to be wrong: to say which assumptions produce which regimes, which conclusions survive a change of functional form, and what observation would count against them. The most useful next steps are a population-level treatment that can be set against grabby-aliens constraints, a collapse term that produces trajectories rather than flags, and a nondimensionalisation that reduces the parameter count to the few groups that actually govern the behaviour.

Data and code availability

The simulator, the verification harness and the scripts that regenerate every figure in this paper are available at https://github.com/RehmozAyub/Ulm_QH_recursive_observability_explorer. A browser implementation requiring no installation is included. *[Mint a Zenodo DOI before submission.]*

Acknowledgements

This work began at the Ulm University Quantum Hackathon, June 2026.

A Notation

Symbol	Type	Meaning
τ	—	dimensionless time
I	state	effective capability
R	state	regulatory quality index, $R \in [0,1]$
O	state	external observability, $O \geq O_{\text{floor}}$
A	driver	ordinary amplification pressure
A_{rec}	driver	recursive amplification
A_{ref}	driver	reflective amplification
Q	driver	institutional quality
a, b, c, s_I	coeff.	ordinary growth, recursive growth, regulatory damping, saturation
u, v, w, s_R	coeff.	reflective learning, institutional gain, erosion, decay
p, q, r	coeff.	energy-use, broadcast and expansion weights
m, n	coeff.	compression and stealth suppression *rates*
$F(I)$	function	recursion kernel
$\Pi(I)$	function	total signature production, $pE + qB_{\text{c}} + rX$
$\Sigma(I,R)$	function	total fractional suppression rate, $mC + nS$
$h(O)$	function	detection map, $1 - e^{-\kappa O}$
B, W	derived	regulation building and erosion aggregates, Eq. (6)
R_∞	derived	regulation equilibrium, $B/(B+W)$
O^*	derived	quasi-steady observability, Eq. (9)
C_R	derived	auxiliary regulation capacity, Eq. (16)
I_{crit}	derived	auxiliary Lambert boundary, Eq. (17)
$N_{\text{true}}, N_{\text{detectable}}, N_{\text{obs}}$	counts	existing, detectable, observed civilizations
$P_{\text{surv}}, P_{\text{det}}, P_{\text{search}}$	prob.	survival, detection, search coverage
κ	param	detection sensitivity
$O_{\text{floor}}, O_{\text{detectable}}$	param	thermodynamic floor; detection threshold
$I_{\text{advanced}}, R_{\text{min}}, \Theta$	param	advanced-capability, regulatory and runaway thresholds
$\eta, \mathcal{E}$	param	regulation exponent; alignment factor (Lambert case)
$W(\cdot)$	function	Lambert W, principal branch

B Closed-form solutions

Two exact solutions exist and are used as verification oracles in §6.

B.1 Regulation

With constant drivers, Eq. (6) is linear in R and independent of I and O:

$$\frac{\mathrm{d}R}{\mathrm{d}\tau} = B - (B + W)\, R, \qquad B = uA_{\mathrm{ref}} + vQ, \quad W = wA_{\mathrm{rec}} + s_R, \tag{19}$$

so that

$$R(\tau) \;=\; R_\infty + \big(R_0 - R_\infty\big)\, e^{-(B+W)\tau}, \qquad R_\infty = \frac{B}{B+W}. \tag{20}$$

Because $B, W \geq 0$ we have $R_\infty \in [0, 1]$ immediately, which is the substance of Proposition 1. Equation (20) also gives the relaxation time $1/(B+W)$: regulation responds faster when either building or erosion is strong.

B.2 Capability in the reduced stage models

For stages 0–2 the regulatory damping term is inactive and recursive amplification is absent, so Eq. (5) reduces to the logistic equation

$$\frac{\mathrm{d}I}{\mathrm{d}\tau} = aA\, I - s_I I^2, \tag{21}$$

with solution

$$I(\tau) = \frac{K_I}{1 + \left(\dfrac{K_I}{I_0} - 1\right) e^{-aA\tau}}, \qquad K_I = \frac{aA}{s_I}. \tag{22}$$

With the stage-1 parameters ($a = 0.30$, $A = 1$, $s_I = 0.10$, $\lambda = 0.30$) this gives $K_I = 3.0$ exactly, and hence a limiting observability under model A of

$$O_\infty = 1 - e^{-\lambda K_I} = 1 - e^{-0.9} = 0.5934. \tag{23}$$

Both are reproduced by the solver to four decimal places (Table 5), which is a stringent check because it tests the integrator, the parameter plumbing and the stage configuration simultaneously.

B.3 The general Bernoulli form

More generally, whenever recursive amplification is absent ($bA_{\mathrm{rec}} = 0$), Eq. (5) is a Bernoulli equation $\mathrm{d}I/\mathrm{d}\tau = g(\tau)I - s_I I^2$ with $g(\tau) = aA - cR(\tau)$. The substitution $z = 1/I$ linearises it to $z' + gz = s_I$, giving

$$I(\tau) = \left[e^{-G(\tau)} \left(\frac{1}{I_0} + s_I \int_0^\tau e^{G(s)}\, \mathrm{d}s\right)\right]^{-1}, \qquad G(\tau) = \int_0^\tau g(s)\, \mathrm{d}s, \tag{24}$$

where G is available in closed form because R is (Eq. (20)). This provides a quadrature-based check independent of the ODE solver.

C Parameters

Table 7 gives the default values and admissible ranges used throughout. All quantities are dimensionless. None is calibrated against data (§9); the ranges bound exploration rather than expressing prior belief.

Stage configurations differ from these defaults only in the drivers, the functional selections of Table 2, and the initial conditions. Stages 0–2 additionally disable the regulatory damping term, reproducing the reduced models of the source framework literally rather than approximating them by parameter choice.

D Verification protocol

D.1 Configuration matrix

All oracles in §6 are evaluated over the same twenty configurations: the eight stage presets; five configurations exercising each recursion kernel of Table 2; four exercising each static observability model; one with time-varying drivers; one Lambert configuration; and one selecting the non-default variant of every observability component. The intent is coverage of the registry rather than of parameter space, which is addressed separately by the randomised testing of oracle E.

D.2 Property-based testing

Oracle E draws parameters uniformly from the ranges in Table 7, together with a randomly selected recursion kernel and randomly selected suppression forms, and asserts

$$O(\tau) \geq 0 \quad \text{and} \quad 0 \leq R(\tau) \leq 1 \qquad \text{for all sampled } \tau. \tag{25}$$

Three hundred draws produced no violations, with global extrema $\min O = 2.16 \times 10^{-3}$ and $R \in [4.0 \times 10^{-3}, 0.999]$.

This is a stronger form of evidence than checking the presets, because the propositions assert structural properties: a bound that held only for tuned configurations would not be a bound at all. The randomised search restricts the recursion kernel to the non-explosive variants, since the blow-up kernels terminate on the runaway guard and would test the guard rather than the invariant.

D.3 Cross-implementation comparison

The Python and JavaScript implementations share no numerical machinery. The JavaScript version implements its own Dormand–Prince tableau, its own quartic dense-output interpolation, its own initial-step heuristic, and a Lambert W_0 by Halley iteration rather than by library call. Trajectories are compared pointwise at 51 sampled times per configuration, and regime labels are compared directly.

Reported discrepancies are therefore attributable to independent accumulations of truncation error at the same requested tolerance, and their magnitude (5.2×10^{-5} in I, against a requested relative tolerance of 10^{-6} over integration horizons of $\tau \leq 100$) is consistent with that interpretation.

D.4 Reproduction

```
# Python reference: 62 tests
cd recursive_observability_explorer
PYTHONPATH=. python -m pytest tests -q
```

```
# Regenerate every data figure in this paper
cd paper
node js_dump.js          # browser-implementation trajectories
python make_figures.py   # figures 5-10

# Build the paper
latexmk -pdf main.tex
```

Environment used: Python 3.11 with NumPy 2.4.2 and SciPy 1.17.1 (Harris et al., 2020; Virtanen et al., 2020); Node.js 24.11 for the browser implementation. Figures 1–4 are TikZ and require no external data.

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Table 7: Default parameter values and admissible ranges.

Symbol	Default	Range	Role
Capability, Eq. (5)			
a	0.30	$[0, 2]$	ordinary growth
b	0.20	$[0, 2]$	recursive growth
c	0.50	$[0, 3]$	regulatory damping
s_I	0.10	$[0, 1]$	capability saturation
Regulation, Eq. (6)			
u	0.40	$[0, 2]$	reflective learning
v	0.30	$[0, 2]$	institutional gain
w	0.50	$[0, 3]$	erosion by recursive amplification
s_R	0.15	$[0, 1]$	regulation decay
Observability, Eq. (8)			
p	0.50	$[0, 3]$	energy-use weight
q	0.30	$[0, 3]$	broadcast weight
r	0.40	$[0, 3]$	expansion weight
m	0.50	$[0, 3]$	compression suppression rate
n	0.30	$[0, 3]$	stealth suppression rate
Drivers			
A, A_{rec}, A_{ref}, Q	1.0	$[0, 5]$	amplification and institutional drivers
Shape and detection			
K	5.0	$[0.1, 50]$	half-saturation of the saturating kernel
λ	0.30	$[0.01, 3]$	observability scale
κ	1.0	$[0.01, 10]$	detection sensitivity
P_{search}	0.50	$[0, 1]$	search coverage
P_{surv}	0.80	$[0, 1]$	survival probability
Thresholds			
Θ	5.0	$[0.1, 50]$	runaway ratio A_{rec}/R
I_{runaway}	50	$[1, 1000]$	runaway capability
$R_{\min}$	0.05	$[0, 1]$	regulatory exhaustion
I_{advanced}	5.0	$[0, 100]$	advanced capability
$O_{\text{detectable}}$	0.10	$[0, 1]$	detection threshold
O_{floor}	10^{-3}	$[0, 0.1]$	thermodynamic floor
Numerics			
$\tau_{\max}$	50	$[1, 500]$	integration horizon
I_{abort}	10^4	—	runaway guard (§6)
rtol / atol	10^{-6} / 10^{-9}	—	solver tolerances